# Smart City Energy Optimization & Prediction

An End-to-End Azure Data Engineering & AI Platform


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# Project Vision

### The Problem

Modern cities face unpredictable energy spikes due to rapid growth. Weather conditions and traffic congestion significantly impact grid stability, leading to inefficiencies.

### The Solution

An automated Azure platform integrating Energy, Weather, and Traffic data. Utilizing Medallion Architecture (Raw → Bronze → Silver → Gold) for robust processing.

### Unique Value

**Proactive Optimization:** Not just reporting, but identifying "Risk Zones" and providing actionable load-shifting recommendations before outages occur.

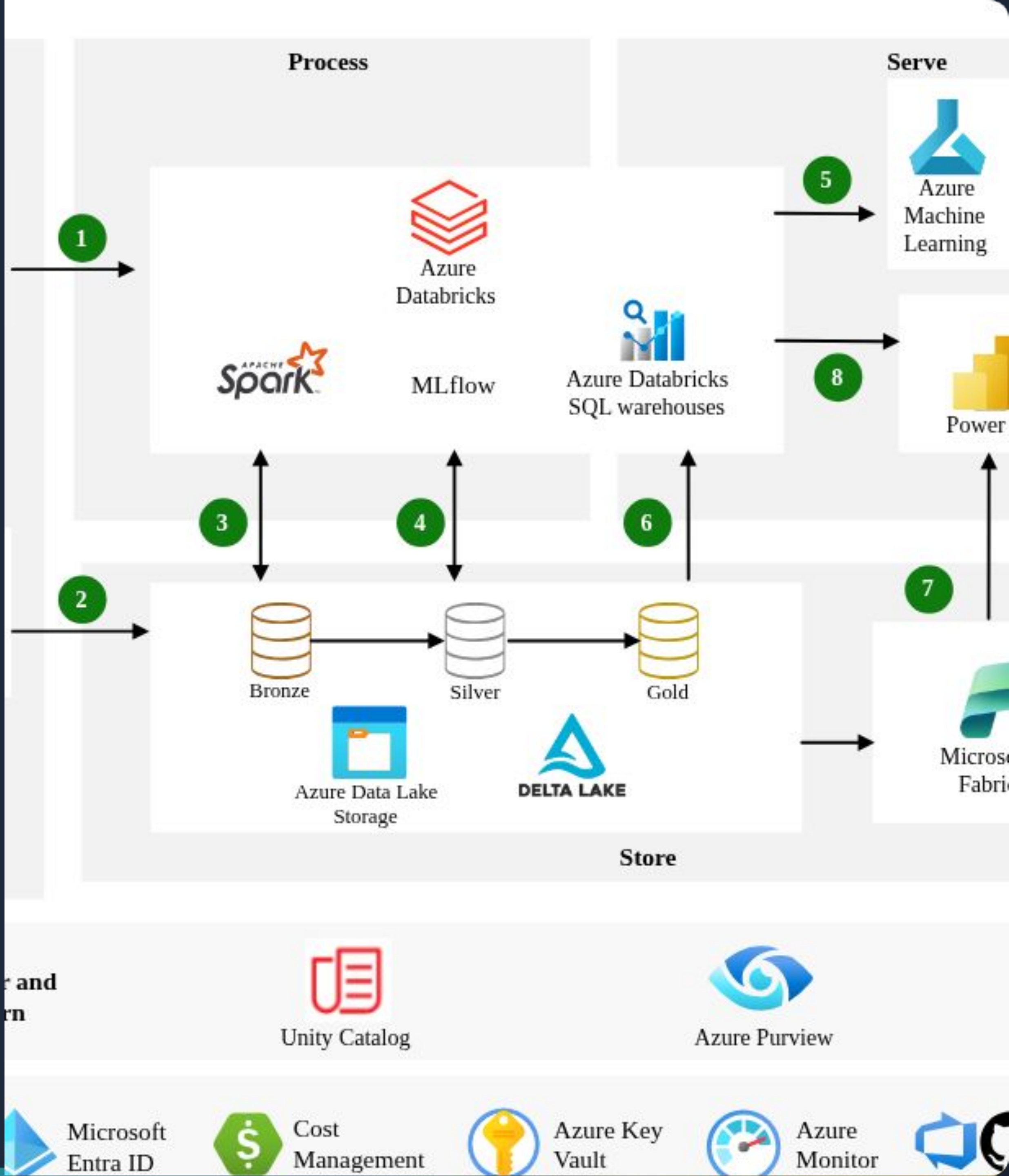
# System Architecture

**Data Sources:** Integration of Batch CSVs and Real-time External APIs (Weather, Traffic).

**Ingestion & Processing:** Azure Data Factory for orchestration and Databricks for transformation.

**AI Modeling:** Azure ML Designer for forecasting and risk classification.

**Visualization:** Synapse SQL Views powering interactive Power BI dashboards.



# Users & Core Features

## Target Audience

**City Planners:** Require long-term insights for sustainable infrastructure development.

**Grid Operators:** Need real-time alerts on peak loads to manage grid stability proactively.

## Key Features

- **Energy Forecasting:** Hourly predictions per region.
- **Risk Identification:** Classifying zones as Low, Medium, or High risk.
- **Optimization Engine:** Smart load balancing recommendations.

# Medallion Data Architecture

A structured flow ensuring data quality and lineage in Azure Data Lake Storage.

### Raw Layer

Stores original data exactly as received (CSVs, JSONs) for audit and backup.

### Bronze Layer

Standardized data with added metadata columns (Load Date, Source System).

### Silver Layer

Cleaned, de-duplicated data with feature engineering (Weather Index, Flags).

### Gold Layer

Aggregated, business-ready data (Forecasts, Risk Scores) for BI & Analytics.

# Languages & Frameworks

## Languages

**Python (PySpark):** Used in Databricks for complex ETL and Feature Engineering.

**SQL:** Used in Synapse Serverless Pools for creating reporting views.

## Core Frameworks

**Azure Data Factory:** Orchestration pipelines.

**Azure Databricks:** Big Data processing.

**Delta Lake:** ACID transactions.

## AI & Analytics

**Azure ML Designer:** Visual ML pipelines.

**MLflow:** Model lifecycle management.

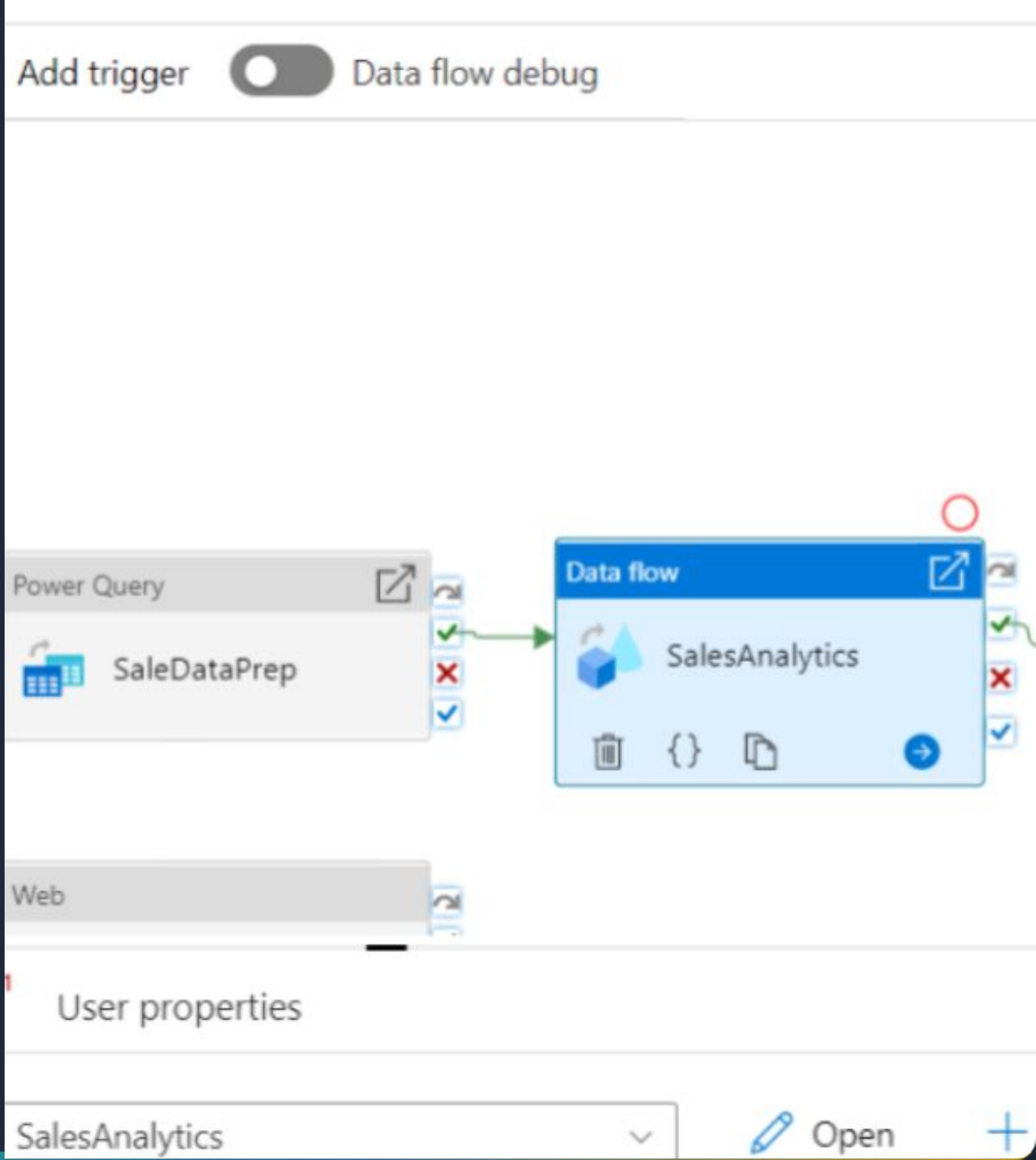
**Power BI:** Interactive visualization.

**Monitoring:** Configured with "Daily Triggers" and Azure Monitor Alerts for failures.

# Ingestion Pipelines

## Presenter: Nagat Osama

- **Pipeline 1 (Energy):** Ingests daily batch CSV files into ADLS Raw and promotes to Bronze.
- **Pipeline 2 (Weather):** Fetches real-time REST API data, handling JSON responses.
- **Pipeline 3 (Traffic):** Captures live congestion data via API for correlating load spikes.



# Data Processing (Silver Layer)

**Presenter: Ebrahim Ateff**

## Data Cleaning

Implemented PySpark logic to remove duplicates and handle missing values using mean imputation. This ensures high-quality input for the ML models.

## Transformation

**Weather:** Exploded complex JSON arrays to align history.

**Traffic:** Flattened nested structures to calculate congestion indices.

**Master Table:** Joined all sources into silver.smartcity_master.

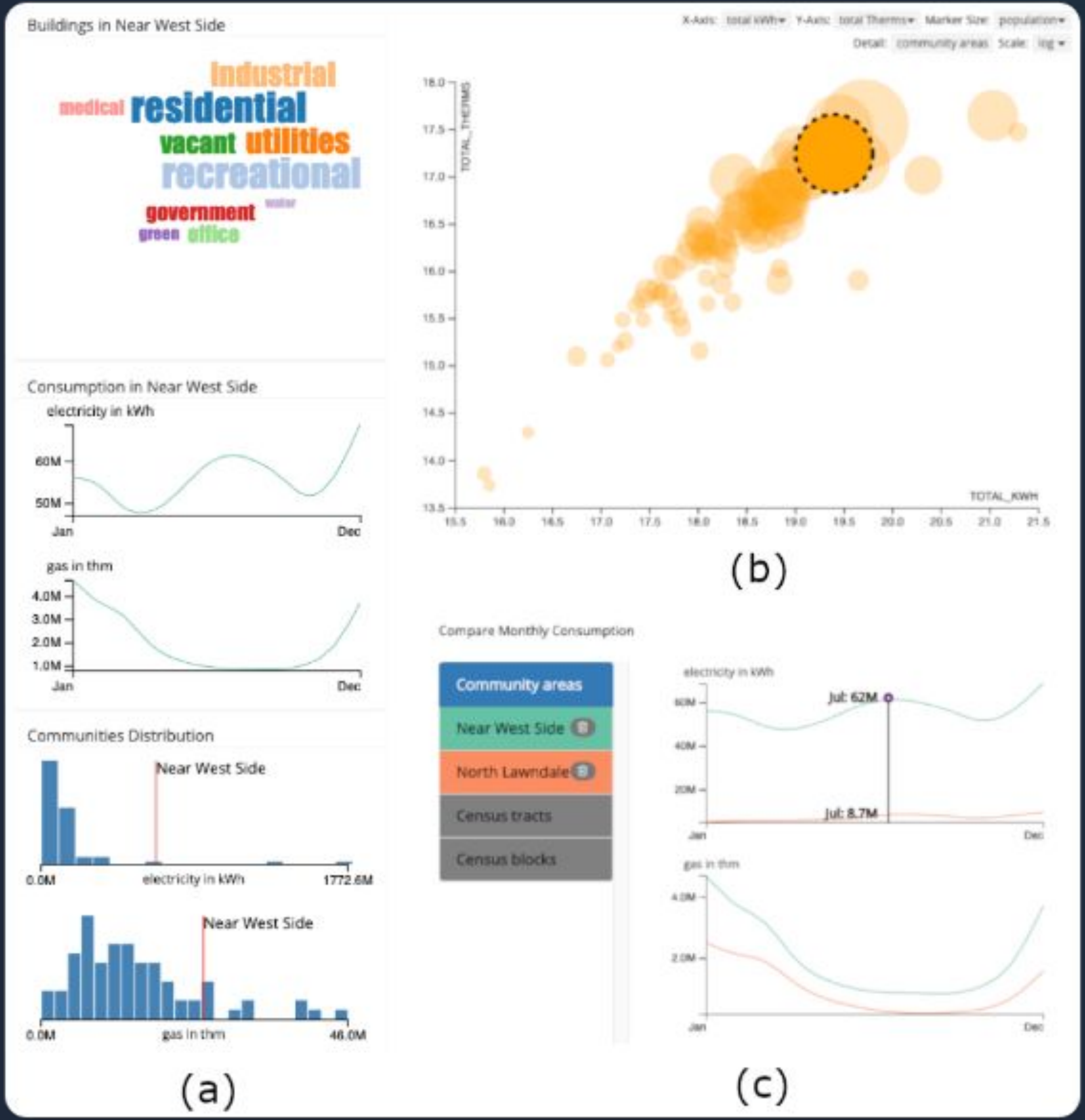
# Forecasting & AI Models

**Presenter: Abdullah Mohamed**

**Data Strategy:** Split historical data into 80% Training and 20% Testing sets for validation.

**Models:** Trained ARIMA and Prophet models on the Silver Master table to predict demand.

**Gold Outputs:**
- gold.energy_forecast: Hourly predictions.
- gold.risk_zones: Risk classification.
- gold.optimization: Load shifting logic.



# Power BI Integration

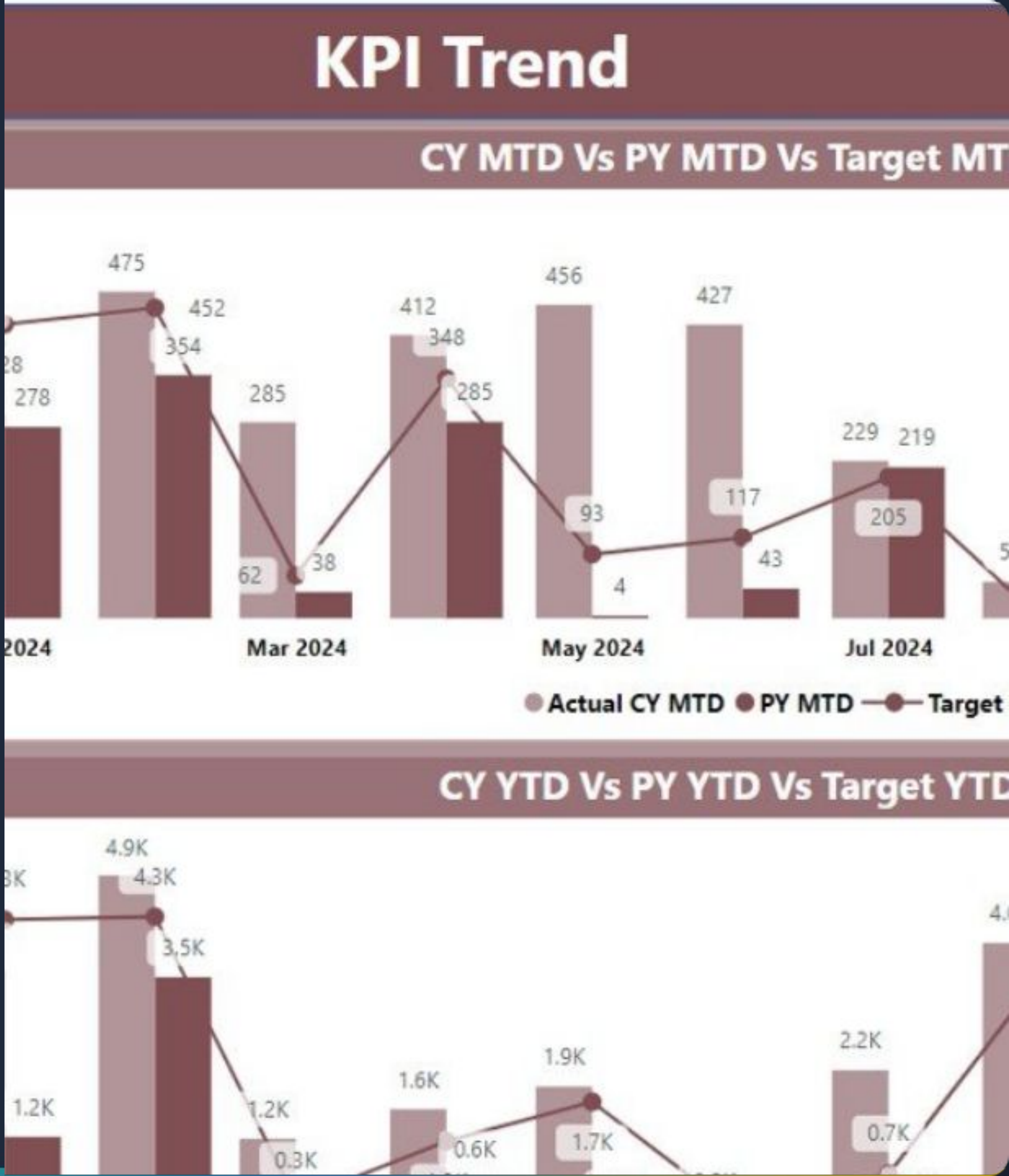
**Presenter: Merna Sameh**

**Synapse Integration:** Leveraged Serverless SQL Views (view_energy_forecast) to expose Gold data securely to Power BI.

**Key Visuals:**

- Regional Energy Risk Analysis map.
- Forecast vs. Actual Consumption trends.

**Impact:** Provides decision-makers with a real-time view of grid health and optimization opportunities.



# Project Deliverables

**Technical Assets:** Fully functional Azure Data Factory pipelines (JSON exports) and Databricks PySpark notebooks for ETL.

**Models:** Trained ML forecasting models registered and tracked in MLflow.

**Reports:** Interactive Power BI dashboard (.pbix) and comprehensive technical documentation with architecture diagrams.

# Project Team

### Nagat Osama

Data Engineer

Ingestion Pipelines

### Ebrahim Ateff

Data Engineer

Processing & Transformation

### Abdullah Mohamed

ML Engineer

AI Models & Forecasting

### Merna Sameh

Team Lead

BI & Visualization

# Questions?

Thank you for your attention.

Team 4 | Smart City Energy Project

# Image Sources

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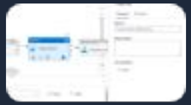

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